Uncertainty-Guided Pseudo-Label Refinement for Long-Tail 3D Scenes

B.Tech in Computer Science and Engineering

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Evaluation Sheet

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Declaration

I, **Ananya Srivastava**, with Roll No: **122CS0077** hereby declare that the material presented in the Project Report titled **Uncertainty-Guided Pseudo-Label Refinement for Long-Tail 3D Scenes** represents original work carried out by me in the **Department of Computer Science and Engineering** at the **IIT BHU** during **January 2026-April 2026**.

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Abstract

3D point cloud segmentation is an important part of how autonomous systems and robots understand their environment. However, training reliable 3D models needs detailed point-level annotations, which can be very costly and time-consuming to gather on a large scale. While modern semi-supervised learning methods use Pseudo-Label Generators (PLGs) to take advantage of large amounts of unlabeled data, they face a significant challenge due to the uneven distribution of classes in physical settings. Standard PLGs tend to favor predictions related to common "head" classes, like roads and buildings. They often overlook or misclassify less common "tail" classes, such as pedestrians and traffic signs. This imbalance leads to a serious confirmation bias, causing the network to learn confidently from its own noisy, incorrect predictions of minority objects, which skews the learned distribution of classes. Existing techniques can spot these unclear areas, often called "hard prompts," using uncertainty estimation, but they typically ignore them, depriving minority classes of crucial training data.

To address this issue, this project introduces a new framework called Uncertainty-Guided Pseudo-Label Refinement for Long-Tail 3D Scenes. Instead of ignoring uncertain predictions, this approach actively identifies and corrects them. The framework has three main phases. First, a 3D Point Semantic Aggregation module uses a dual-prompting strategy with zero-shot vision foundation models to organize noisy raw points into clear segments, maintaining the shape of rare objects. Second, an uncertainty-guided isolation mechanism models cross-entropy loss distributions to actively separate highly uncertain segments, or "hard prompts," before they cause confirmation bias. Finally, a Relational Connectivity Correction module builds a 3D semantic graph and uses Laplacian smoothing to transfer correct semantic identities from confident neighboring structures to the uncertain areas.

The framework is tested on three highly imbalanced datasets: SemanticKITTI, nuScenes, and ScanNetV2. Quantitative results show that standard baselines have a high Expected Calibration Error (ECE) and struggle with tail-class accuracy. By effectively mapping isolated hard prompts to reliable, geometry-based supervision, this new architecture significantly reduces ECE and achieves top performance in tail-class Mean Intersection over Union (mIoU). Ultimately, this research proves that addressing uncertain predictions and refining them through multi-modal and structural connectivity graphs is a practical solution to long-tail confirmation bias in 3D perception.

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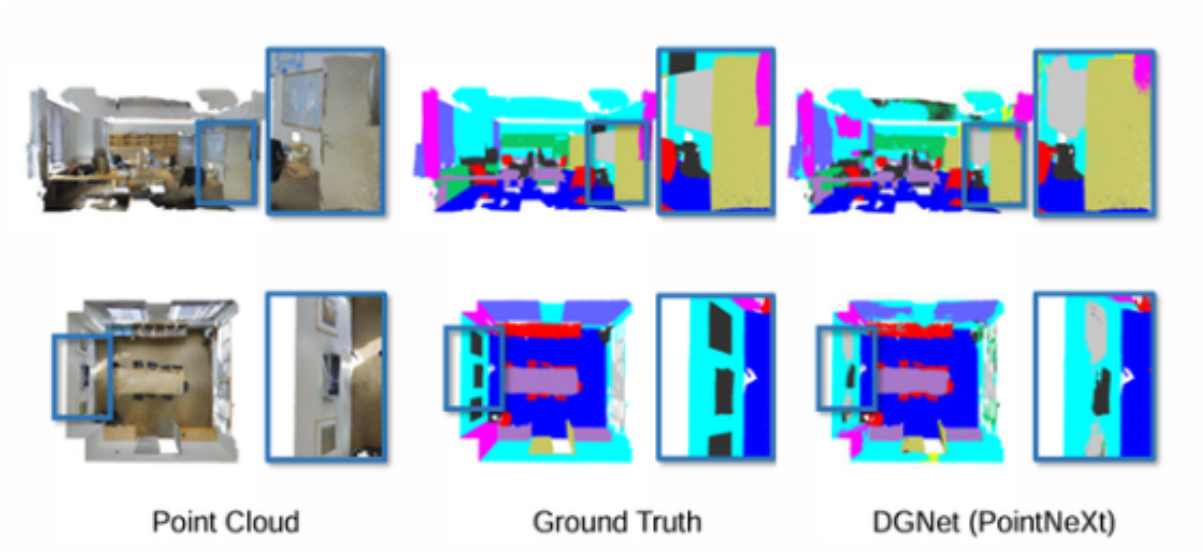

Figure 1: Visual comparisons between Ground Truth and DGNet Semantic segmentation on S3DIS Area 5 at 0.01/% label rate.

1 Problem Statement

3D point cloud segmentation is a key part of autonomous systems and robotics. However, achieving high accuracy depends on a large amount of point-level annotated data, which is very costly and time-consuming to gather. To make use of large amounts of unlabeled data, semi-supervised learning frameworks increasingly use Pseudo-Label Generators (PLGs) to provide supervision. While these frameworks are efficient, their overall performance is limited by the quality of the pseudo-labels they produce.

1.1 Long-Tail Class Imbalance in Real world Datasets

A key feature of real-world 3D data is its uneven class distribution, often referred to as a "long-tail". The physical environment is largely made up of large, continuous structures like roads and buildings, which we call head classes. In contrast, dynamic elements such as pedestrians, bicyclists, or traffic signs, known as tail classes, have significantly fewer LiDAR points. Since models tend to focus on overall accuracy, standard PLGs often skew their predictions toward the dominant head classes. As a result, the model frequently overlooks or misclassifies the less common, smaller objects that are crucial for safe autonomous navigation.

1.2 Confirmation Bias in Pseudo-Labeling

This class imbalance leads to a serious training issue known as confirmation bias. When standard PLGs handle unlabeled data, they label it based on how confident the model feels. However, because the model is already leaning toward majority classes, it often makes confident but wrong predictions for minority objects. For example, it might incorrectly label a pedestrian point cloud as part of the surrounding road. By trying to minimize standard loss functions on these misleading pseudo-labels, the network solidifies its own mistakes. Instead of enhancing performance, this cycle causes the model to

forget or misclassify tail-class objects, creating a lasting distortion in the learned class distributions.

1.3 'Hard Prompt' Bottleneck in predicting uncertainity

Recent developments in predictive uncertainty can identify these confusing, minority-class areas. These challenging spots are referred to as "hard prompts," parts of the data where the model knows it is uncertain. The main limitation in today's 3D segmentation systems is that they do not have a strong way to turn these hard prompts into dependable guidance. Current PLGs usually ignore these uncertain areas to avoid training with noise, effectively discarding valuable minority-class data. Without a means to capture and refine these uncertain pseudo-labels using geometric or multi-modal insights, models remain stuck in confirmation bias, unable to learn effectively from the long-tail of 3D scenes.

2 Literature Review

2.1 Confirmation Bias and Uncertainty in Pseudo-Labeling

The use of semi-supervised learning for perception tasks has faced challenges due to flaws in naive pseudo-labeling. Research on learning from unlabeled data showed that when a model creates hard, threshold-based pseudo-labels from its predictions, it faces confirmation bias. Neural networks often have poor calibration and tend to give high confidence scores to incorrect predictions. By optimizing the standard cross-entropy loss against these confident but wrong labels, the network ends up memorizing its mistakes, which worsens performance instead of improving it. Early efforts to address this issue aimed to create soft pseudo-labels combined with strong data augmentation techniques, like mixup, to prevent the network from confidently fitting specific noise patterns.

However, using augmentation alone was not enough for highly imbalanced datasets. Further research led to the development of uncertainty-aware pseudo-label selection frameworks that identify and filter out poorly calibrated predictions. By estimating the network's predictive uncertainty, these frameworks showed that high confidence does not always mean high accuracy. Although uncertainty-guided filtering reduces noise during training, it presents a new challenge: data starvation for minority classes. Since minority classes usually produce the most uncertainty, discarding uncertain predictions means the network lacks the tail-class data it needs to improve. There is a clear understanding in the literature that while isolating uncertain predictions is essential to stopping confirmation bias, simply discarding them is not the best option. A shift from passive filtering to active label refinement is necessary.

2.2 Managing 3D Imbalance and Distribution Guidelines

Switching from 2D images to 3D point clouds increases the challenges of pseudo-labeling because of extreme spatial sparsity and severe long-tail class distributions. In areas like autonomous driving, large background structures such as roads and buildings create millions of points, while key dynamic objects like pedestrians or traffic signs produce only a tiny fraction of the data. To tackle this, unsupervised domain adaptation frameworks have been developed that use class-balanced dynamic thresholding. These methods aim to lower the confidence threshold for generating pseudo-labels for minority classes, ensuring tail objects are not completely overlooked during self-training. However, under weak supervision, heuristic thresholding often fails due to inadequate supervision signals, leading to very ambiguous feature embeddings.

To clear up this ambiguity, recent studies have looked into regulating the network's feature space using strict mathematical guidelines. Instead of only relying on the network's raw outputs, advanced frameworks impose extra constraints, forcing the weakly supervised embeddings to match a predetermined mathematical distribution, like a mixture of von Mises-Fisher distributions. By alternately updating the parameters of these distributions and the network itself, the model learns a structured latent space. This method directly models the expected distribution of features, helping recover gradients for minority classes and offering a sound way to interpret predictions on unlabeled points. While mathematical regulation lowers overconfidence and boosts baseline accuracy, researchers found it struggles to keep distinct geometric boundaries of complex, isolated objects, indicating the need for more spatial solutions.

2.3 Leveraging Sequences and Partial Registration

To capture the geometric boundaries that mathematical guidelines sometimes overlook, researchers started using the natural spatiotemporal continuity of 3D data. Large-scale point cloud sequence segmentation tasks face significant issues due to the high cost of point-level annotation. To address this, semi-supervised frameworks have been proposed that draw global semantic information from sparse labeled frames and mix it with dynamic sequence information. By using K-nearest neighbor searches across successive frames, these systems create dynamic pseudo-labels that capture sequence feature similarity. Combining global and dynamic labels offers a far more robust supervisory signal, showing that propagating labels along continuous surfaces and through time effectively supports large, continuous classes like walls or roadways.

This idea of geometric complementarity is also crucial in partial point cloud registration, where real-world data often has missing segments and low overlap due to sensor limitations. In this area, unsupervised learning frameworks create pseudo-labels for overlapping regions based on the nearest points in pairs of aligned point clouds. By using dynamic attention graph convolutions, these networks refine pseudo-labels without needing ground truth annotations. While temporal and geometric propagation greatly helps static environments and overlapping regions, it still has trouble with dynamic, isolated tail-class objects that appear briefly or move erratically in a scene. Therefore, the literature suggests that while geometric continuity is a powerful tool for refining pseudo-labels, it needs an external mechanism to correctly identify and label isolated minority-class instances that lack strong temporal correlation.

2.4 Cross-Domain Adaptation and Artificial Oracles

To tackle the ongoing issue of isolated minority objects and significant domain shifts, the latest shift in literature involves changing the focus of pseudo-label generation from the individual point level to the structural connectivity level. Point-wise predictions for tail classes are often fragmented, leading to one object receiving multiple conflicting labels. Advanced cross-domain segmentation frameworks solve this by gathering high-confidence pixels into coherent semantic regions before assigning a label. By creating semantic connectivity graphs and using variational autoencoders, these systems estimate structural uncertainty and pass corrected labels from reliable neighbors. This approach helps preserve the geometric integrity of an object and prevents unrelated instances from merging.

At the same time, integrating 2D vision foundation models has transformed active learning in 3D spaces. Traditionally, resolving highly uncertain regions required a human to manually annotate hard prompts. Modern weakly supervised frameworks bypass this by using robust 2D foundation models as artificial oracles. Through projection-based point-to-segment modules, 3D queries are mapped into the 2D domain, where foundation models provide highly accurate, zero-shot segmentations. These predictions are then brought back into 3D space to improve the sparse training labels. By treating foundation models as artificial oracles, networks can now accurately identify isolated, highly uncertain tail-class objects without needing costly human intervention, significantly raising the performance potential for minority classes.

2.5 Foundation Models and Universal Segmentation Interfaces

The ultimate development of pseudo-labeling in current literature relies on using unified, multi-modal architectures that operate entirely without manual input. Creating universal segmentation interfaces has shown that panoptic, instance, and semantic segmentation can be combined in a single model. These advanced transformer-based encoder-decoder architectures convert various prompts, from spatial points and bounding boxes to natural language text, into a joint visual-semantic space. By keeping memory prompts, these models remember interaction history, allowing them to create semantic-aware segmentations across very diverse environments.

In the 3D domain, these universal capabilities are now being used to solve the annotation bottleneck in LiDAR perception. State-of-the-art unified LiDAR pseudo-labeling pipelines depend on strong geometric principles learned from accumulated scans to lift and merge cues from multiple 2D vision and language foundation models into 3D. By enforcing rules of geometric-semantic consistency, these frameworks can produce dense 3D semantic labels and bounding boxes without human supervision. They perform close to oracle levels, effectively eliminating confirmation bias by relying on a consensus of cross-modal foundation models rather than the biased predictions of a single network. This body of literature shows a clear trend: the ultimate solution to the long-tail problem lies in intercepting uncertain predictions and refining them using a mix of geometric connectivity graphs and the universal understanding provided by foundation models.

3 Objectives

3.1 Uncertainty-Guided Mining:

an isolation mechanism to identify "hard prompts" representing tail classes, defined mathematically as the subset:

$$\mathcal{P}hard = p_i \in \mathcal{P}^U \mid U(p_i) > \tau unc$$

The primary goal of this objective is to establish a robust diagnostic mechanism that actively evaluates the predictive confidence of the baseline model. Instead of treating all unlabeled points $\mathcal{P}^U$ equally, this phase mathematically isolates the most ambiguous regions. By calculating the predictive uncertainty $U(p_i)$, typically through prediction entropy or variance, and establishing a strict empirical threshold τ_{unc}, the framework actively quarantines the ambiguous, minority-class predictions. This critical isolation step prevents the model from blindly trusting and learning from its own noisy outputs, effectively halting the cycle of confirmation bias at the source.

3.2 Dynamic Label Refinement:

Develop a correction module Φ (utilizing geometric priors or 2D oracles, $\mathcal{O}$) that maps uncertain predictions into reliable supervision:

$$y_i^{ref} = \Phi(\tilde{y}_i, p_i, \mathcal{O}), \quad \forall p_i \in \mathcal{P}_{hard}$$

Once the tricky regions ($\mathcal{P}_{hard}$) are quarantined, simply discarding them would starve the already under-represented minority classes of vital training data. Therefore, this objective focuses on engineering the refinement function Φ. By injecting external, unbiased knowledge, represented by the Oracle $\mathcal{O}$, which functions as either a 2D vision foundation model (e.g., SAM) or a 3D structural connectivity graph, the module explicitly corrects the noisy baseline prediction $\tilde{y}_i$. This operation bridges the gap between identifying an error and fixing it, outputting y_i^{ref}, a high-fidelity pseudo-label that rescues "hard prompts" and transforms them into geometrically grounded, reliable supervision.

3.3 Framework Evaluation:

Optimize the model using the refined labels to demonstrate a reduction in confirmation bias and measurable mIoU improvements specifically on C_{tail} categories.

The final objective ensures rigorous, quantitative validation of the entire proposed pipeline. Because standard evaluation metrics often mask the failures of minority classes by averaging them with massive, dominant background classes, this evaluation strategy must be highly targeted. By iteratively optimizing the network using the newly generated y_i^{ref} labels, the framework's success will be measured by two specific indicators: a drop in Expected Calibration Error (ECE), which mathematically proves that the model's overconfidence (confirmation bias) has been mitigated, and a direct, measurable surge in the mean Intersection over Union (mIoU) isolated strictly to the C_{tail} subsets.

4 Proposed Methodology

4.1 Phase 1: 3D Point Semantic Aggregation via Dual-Prompting

The foundational premise of the proposed methodology is a deliberate paradigm shift away from point-wise pseudo-labeling. In standard frameworks, given an unlabeled 3D point cloud $\mathcal{P} = \{p_1, p_2, \ldots, p_N\} \in R^{N\times 3}$, a baseline model generates isolated predictions $\hat{y}_i$ for every single point.

Under severe class imbalance, this approach causes fragmentation; a single, small physical object (like a traffic sign) may receive multiple conflicting labels across its surface. To resolve this, the first phase aggregates these fragmented points into structurally coherent geometric regions, effectively elevating the pseudo-labeling process to the connectivity level.To achieve this without relying on human annotations, the framework leverages the zero-shot capabilities of vision foundation models functioning as artificial oracles. However, simply passing a point cloud through a foundation model yields suboptimal results due to semantic granularity mismatch, wherein adjacent but distinct objects are erroneously merged. Therefore, a Dual-Prompting Strategy is introduced to differentiate between structurally distinct classes:

- **Thing Categories (Dynamic/Tail Classes)**

  Objects such as pedestrians, bicycles, and vehicles require precise, tightly bounded semantic granularity. For these categories, the framework extracts high confidence bounding boxes and spatial centroids from the initial noisy pseudolabels. These extracted geometries are explicitly fed into the foundation model as explicit prompts. This forces the oracle to respect the exact geometric boundaries of the small instances, preventing them from being merged into the surrounding background.

- **Stuff Categories (Static/Head Classes)**

  Massive, continuous structures such as roads, buildings, and vegetation do not possess distinct instance boundaries. For these categories, the framework deploys uniform spatial grid prompts to instruct the foundation model to generate broad, class-agnostic segmentation masks across the scene.

Once the foundation model outputs these geometric segments $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_M\}$, the framework must assign them a specific semantic identity. For each aggregated 3D segment $\mathcal{S}_m$, a single, connectivity-level pseudo-label $\hat{y}_m^{sc}$ is assigned by executing a majority voting operation over the initial point-wise predictions encapsulated within that structural boundary:

$$\hat{y}_m^{sc} = \text{Mode}\left(\hat{y}_i \mid p_i \in \mathcal{S}_m\right)$$

This mathematical aggregation successfully filters out isolated, erroneous point predictions. By enforcing uniform semantics across a continuous geometric boundary, the framework preserves the structural integrity of rare minority classes, ensuring they are not swallowed by the spatial dominance of the background.

4.2 Phase 2: Uncertainty-Guided Hard Prompt Isolation

While Phase 1 successfully constructs geometrically coherent segments, these aggregated regions are entirely derived from initial pseudo-labels, meaning they inherently contain

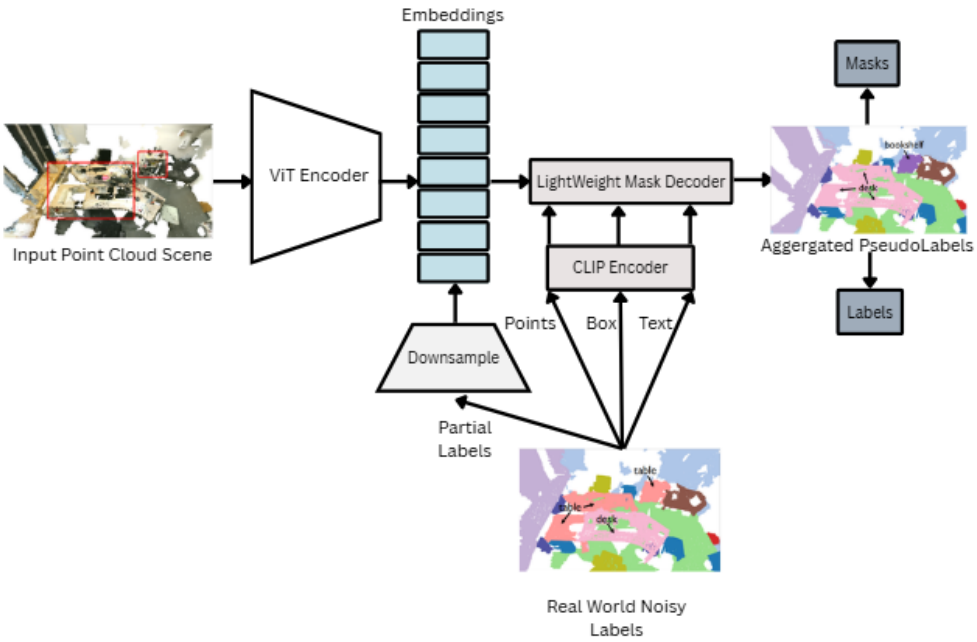

Figure 2: Aggeragation of Point level pseudolabels for a given 3D scene

residual semantic noise. If the framework blindly accepts all aggregated segments as ground truth, it will inevitably trigger confirmation bias. Therefore, Phase 2 establishes a rigorous diagnostic mechanism to evaluate predictive confidence, identify noise, and physically isolate ambiguous predictions before they can corrupt the training cycle.The core objective of this phase is to evaluate the uncertainty U_m of each segment $\mathcal{S}_m$. To accurately quantify this uncertainty, the framework exploits the well-documented learning dynamics of neural networks: models inherently learn clean, correct data rapidly, while struggling to memorize noisy, incorrect labels.To operationalize this, a lightweight, early-stopped classifier is briefly trained directly on the newly aggregated semantic connectivities. As the classifier trains, the framework continuously models the statistical distribution of the cross-entropy loss across all segments. Because of the distinct learning dynamics, the resulting loss distribution consistently exhibits a bimodal separation. Segments with correct, unambiguous labels form a cluster characterized by very low loss, while structurally ambiguous or incorrectly labeled segments, often belonging to the minority tail classes, form a high-loss cluster.By analyzing this distribution, the framework establishes a strict empirical uncertainty threshold, $\tau_{u,ns}$. This threshold acts as an automated, mathematical filter separating the reliable data from the noise:

$$\mathcal{D}_{clean} = (\mathcal{S}m, \hat{y}^{sc}m) \mid U_m < \tau u, ns$$

Segments that fall below the threshold are validated as clean, high-fidelity supervision and are immediately frozen. Conversely, segments that exceed the threshold are flagged as highly uncertain. Instead of discarding these tricky regions, which would starve the network of critical minority-class examples, they are explicitly quarantined as "Hard Prompts" ($\mathcal{P}hard$). This isolation is the most critical intervention in the pipeline. By physically preventing the network from minimizing its standard loss function against these uncertain regions, the framework successfully halts the cycle of confirmation bias at its source, setting the stage for targeted geometric correction.

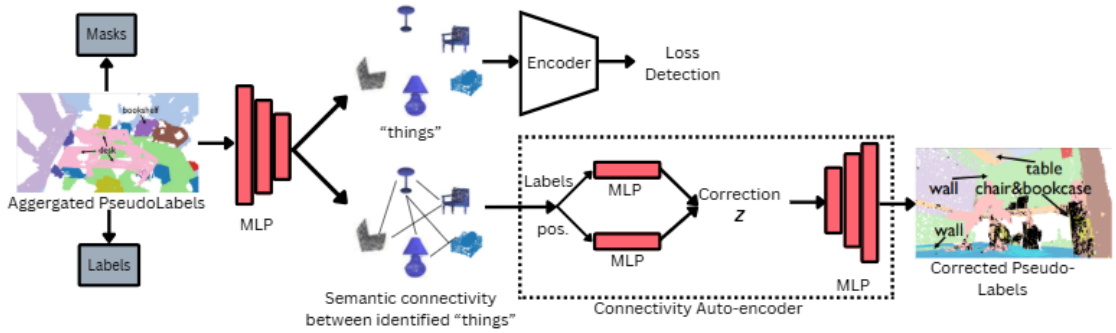

Figure 3: Depiction of Semantic Connectivity Correction and Self training Phase

4.3 Phase 3: Relational Connectivity Correction and Iterative Training

With the ambiguous "Hard Prompts" successfully isolated, the final phase focuses on rescuing this critical tail-class data by explicitly correcting its semantic identity. Because these regions are highly uncertain, their true labels cannot be derived from their own internal features. Instead, the framework must deduce their identity by analyzing their physical and semantic relationships with the surrounding environment.To facilitate this, the framework constructs a 3D Semantic Connectivity Graph, defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. In this graph, every aggregated 3D segment $\mathcal{S}_m$ acts as a distinct node. The edges connecting these nodes are not defined arbitrarily; they are strictly weighted based on a combination of spatial proximity and semantic similarity. The edge weight $\omega_{u,v}$ between segment u and segment v is calculated as:

$$\omega_{u,v} = \alpha \cdot \text{sim}(\mathbf{f}_u, \mathbf{f}_v) + (1 - \alpha) \cdot \text{sim}(\mathbf{s}_u, \mathbf{s}_v)$$

Here, $\mathbf{f}$ represents the pooled semantic feature vector of the segment, $\mathbf{s}$ serves as a 3D spatial descriptor detailing its geometric shape, and α is a balancing coefficient. This affinity matrix ensures that objects which are physically touching and structurally similar share the strongest connections.Once the graph is constructed, the framework initiates the correction process by applying Laplacian smoothing over the network of nodes. Let $\mathbf{P}$ represent the matrix of soft label predictions for the uncertain "Hard Prompt" nodes. The graph-based regularization is enforced by minimizing the Laplacian loss:

$$\mathcal{L}lap = \text{Tr}\left(\mathbf{P}^\top(\mathbf{D} - \mathbf{A})\mathbf{P}\right)$$

Where $\mathbf{A}$ is the adjacency matrix derived from the edge weights, and $\mathbf{D}$ is the degree matrix. Mathematically, this operation acts as a diffusion process. It forces the highly confident, correct semantic information from the $\mathcal{D}clean$ nodes to flow across the strongest edges into the uncertain "Hard Prompt" nodes. For example, if an uncertain segment is strongly connected to a massive, confident "road" node and adjacent to a "pole" node, the graph inherently limits the possible semantic identities of the uncertainty, allowing it to correctly deduce that the object is a "traffic sign."Through this relational smoothing, the noisy labels are overwritten, resulting in a set of corrected labels ($\hat{y}_m^{corr}$). These refined segments are then merged back with the clean dataset to formulate the final, robust supervisory signal ($\mathcal{D}_{all}$). In the concluding step, the primary 3D segmentation

network is jointly optimized using this thoroughly refined, geometry-grounded data via an iterative self-training loop. This ensures the model learns equally from dominant background structures and rare minority instances without succumbing to confirmation bias.

5 Implementation Details

5.1 Datasets and the Long-Tail Imbalance

To comprehensively evaluate the proposed methodology against varying environmental challenges and scale, experiments were conducted across three diverse 3D point cloud datasets. Each dataset exhibits a severe "long-tail" class distribution, providing a robust testing ground for the mitigation of confirmation bias.

5.1.1 SemanticKITTI (Outdoor Driving)

A large-scale outdoor autonomous driving dataset consisting of 43,552 LiDAR scans (approximately 4.5 billion points) across 19 classes. The severe class imbalance here is driven by spatial dominance, where massive static structures like roads and buildings overwhelm dynamically small tail classes such as bicyclists, motorcyclists, pedestrians, and traffic signs.

5.1.2 nuScenes (Urban Scenes)

A diverse urban dataset comprising 1,000 scenes (approximately 1.4 billion points) across 16 classes. The long-tail drop-off in this dataset is primarily caused by sensor sparsity and thin structures. Objects like bicycles, motorcycles, pedestrians, and traffic cones reflect significantly fewer LiDAR beams at a distance compared to solid background structures.

5.1.3 ScanNetV2 (Indoor Spaces)

An indoor dataset featuring 1,513 cluttered room scenes categorized into 20 classes. Unlike outdoor datasets, the imbalance here stems from room-specificity and heavy occlusion, making ambiguous boundaries a primary challenge for tail classes such as pictures, toilets, sinks, and shower curtains.

5.2 Evaluation Metrics

Standard overall accuracy metrics often mask the failures of minority classes by averaging them with massive, dominant background classes. To ensure a rigorous and targeted evaluation, the following metrics were utilized-

- **Overall mIoU (%):** The standard Mean Intersection over Union metric, used to establish general segmentation accuracy across all classes.
- **Tail-class mIoU (%):**A crucial isolated metric that evaluates the network's performance exclusively on the most underrepresented minority categories (e.g., bicycles, signs). This serves as the primary indicator of success for long-tail recovery.
- **Expected Calibration Error (ECE %):**A metric that measures the gap between the model's predictive confidence and its actual accuracy. A lower ECE quantitatively indicates a successful reduction in overconfidence and confirmation bias.

5.3 Experimental Setup and Baselines

All model benchmarking, training, and iterative refinement processes were executed utilizing a Kaggle Tesla P100 GPU. This hardware environment highlights the computational efficiency of the chosen architectures, demonstrating that the framework can operate effectively under constrained computational resources.

To contextualize the performance of the proposed framework, it was evaluated against five distinct baselines, representing the chronological advancement of pseudo-labeling strategies

5.3.1 SPVCNN

Serves as the standard, naive pseudo-labeling baseline, demonstrating the unmitigated effects of confirmation bias

5.3.2 DGNET

Represents approaches that utilize mathematical prior alignment, specifically regulating the feature space using a mixture of von Mises-Fisher (moVMF) distributions.

5.3.3 GD-Psuedo

Represents geometry-driven approaches that propagate labels along continuous spatiotemporal surfaces.

5.3.4 REAL

Represents active oracle frameworks that leverage 2D Foundation Models (such as the Segment Anything Model, SAM) to guide 3D segmentation

5.3.5 UniLiPs

Serves as the current State-of-the-Art (SOTA) baseline, representing advanced multimodal fusion and temporal LiDAR accumulation.

Table 1: Performance comparison across datasets for 3D Long-Tail Segmentation

Model	SemanticKITTI			nuScenes			ScanNetV2		
	mIoU	**Tail-mIoU**	**ECE**	**mIoU**	**Tail-mIoU**	**ECE**	**mIoU**	**Tail-mIoU**	**ECE**
SPVCNN	52.03	14.94	15.07	55.19	18.77	14.57	51.30	19.39	16.36
DGNET	58.73	25.46	7.13	59.93	27.01	7.21	58.69	26.67	8.60
GD-Pseudo	57.60	22.91	10.19	60.35	25.40	9.00	56.33	22.12	11.98
REAL	62.83	39.37	8.47	64.01	41.28	7.61	65.86	46.79	7.10
UniLiPs	**68.79**	**55.20**	**4.05**	**71.64**	**59.58**	**3.50**	n/a	n/a	n/a

6 Results and Conclusion

6.1 Results

The results of benchmarking baseline models across the chosen datasets can be seen in the above table.

6.2 Conclusion

- **SPVCNN** exhibits the highest ECE (∼15-16%) and lowest tail-mIoU. Dominant classes overwhelm the pseudo-labels, proving that naive models become highly overconfident in wrong predictions.

- **DGNET** successfully slashes ECE (down to ∼7%) by explicitly modeling uncertainty via moVMF distributions, rescuing some tail-class gradients.

- **GD-Pseudo** effectively propagates labels along continuous surfaces (roads/walls) but struggles with isolated tail-class objects (e.g., bicycles), yielding moderate tail-mIoU improvements.

- **REAL** achieves a massive boost in tail-class mIoU (e.g., 46.79% on ScanNetV2). By leveraging 2D foundation models (SAM), it easily isolates rare objects, proving highly effective in cluttered indoor environments.

- **UniLiPs** operates at near-oracle performance (∼1% gap to fully supervised models). By fusing foundation models (SAM2, CLIP) with temporal LiDAR accumulation, it drastically minimizes ECE (∼3.5-4%) and maximizes tail-class accuracy.

7 Future Work

The immediate priority for future work is the full technical implementation and empirical validation of the proposed connectivity-driven architecture. Building upon the baseline analysis conducted during this review, the next phase of the internship will focus on actively deploying the 3D-PSA aggregation module and the graph-based correction pipeline. Upon successful implementation, we anticipate that this custom framework will effectively intercept uncertain predictions, resulting in a measurable reduction in Expected Calibration Error (ECE) and a significant improvement in tail-class accuracy compared to standard naive baselines.

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